

Tutorial 4 – Solutions

Iterative methods for solving systems of linear equations

Exercise 1.1(a) – Find N

From

$$A = M - N$$

we get

$$N = M - A.$$

So

$$N = \begin{pmatrix} 2 & 1 & 0 & 0 \\ 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 1 & 2 \end{pmatrix} - \begin{pmatrix} 2 & 1 & \frac{1}{2} & 1 \\ 1 & 1 & 0 & \frac{1}{2} \\ \frac{1}{2} & 0 & 1 & 1 \\ 1 & \frac{1}{2} & 1 & 2 \end{pmatrix} = \begin{pmatrix} 0 & 0 & -\frac{1}{2} & -1 \\ 0 & 0 & 0 & -\frac{1}{2} \\ -\frac{1}{2} & 0 & 0 & 0 \\ -1 & -\frac{1}{2} & 0 & 0 \end{pmatrix}.$$

The iterative method is

$$x^{(k+1)} = Gx^{(k)} + c, \quad G = M^{-1}N, \quad c = M^{-1}b.$$

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Exercise 1.1(a) – Compute M^{-1} by Gauss elimination

We compute the inverse from the augmented matrix

$$\left(\begin{array}{cccc|cccc} 2 & 1 & 0 & 0 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 & 0 & 0 & 1 & 0 \\ 0 & 0 & 1 & 2 & 0 & 0 & 0 & 1 \end{array} \right).$$

$$R_1 \leftrightarrow R_2 \quad \left(\begin{array}{cccc|cccc} 1 & 1 & 0 & 0 & 0 & 1 & 0 & 0 \\ 2 & 1 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 1 & 0 & 0 & 1 & 0 \\ 0 & 0 & 1 & 2 & 0 & 0 & 0 & 1 \end{array} \right).$$

$$R_2 \leftarrow R_2 - 2R_1, \quad R_4 \leftarrow R_4 - R_3 \quad \left(\begin{array}{cccc|cccc} 1 & 1 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & -1 & 0 & 0 & 1 & -2 & 0 & 0 \\ 0 & 0 & 1 & 1 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & -1 & 1 \end{array} \right).$$

$$R_2 \leftarrow -R_2, \quad R_1 \leftarrow R_1 - R_2, \quad R_3 \leftarrow R_3 - R_4$$

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Therefore

$$M^{-1} = \begin{pmatrix} 1 & -1 & 0 & 0 \\ -1 & 2 & 0 & 0 \\ 0 & 0 & 2 & -1 \\ 0 & 0 & -1 & 1 \end{pmatrix}.$$

Exercise 1.1(a) – Compute G and c

We compute

$$G = M^{-1}N, \quad c = M^{-1}b.$$

Hence

$$G = \begin{pmatrix} 0 & 0 & -\frac{1}{2} & -\frac{1}{2} \\ 0 & 0 & \frac{1}{2} & 0 \\ 0 & \frac{1}{2} & 0 & 0 \\ -\frac{1}{2} & -\frac{1}{2} & 0 & 0 \end{pmatrix}, \quad c = \begin{pmatrix} 1 \\ -1 \\ 0 \\ 0 \end{pmatrix}.$$

So Method 1 is

$$x^{(k+1)} = Gx^{(k)} + c.$$

Componentwise,

$$\begin{cases} x_1^{(k+1)} = 1 - \frac{1}{2}x_3^{(k)} - \frac{1}{2}x_4^{(k)}, \\ x_2^{(k+1)} = -1 + \frac{1}{2}x_3^{(k)}, \\ x_3^{(k+1)} = \frac{1}{2}x_2^{(k)}, \\ x_4^{(k+1)} = -\frac{1}{2}x_1^{(k)} - \frac{1}{2}x_2^{(k)}. \end{cases}$$

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Exercise 1.1(b) – $\|G\|_1$ and $\|G\|_\infty$

For the 1-norm, we sum absolute values by columns:

$$\text{col sums of } |G| : \quad \frac{1}{2}, 1, 1, \frac{1}{2}.$$

Therefore

$$\|G\|_1 = 1.$$

For the infinity norm, we sum absolute values by rows:

$$\text{row sums of } |G| : \quad 1, \frac{1}{2}, \frac{1}{2}, 1.$$

Therefore

$$\|G\|_\infty = 1.$$

These two norms do *not* allow us to conclude convergence directly.

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These two norms do *not* allow us to conclude convergence directly.

Exercise 1.1(b) – Detailed computation of $G^T G$

We use

$$\|G\|_2 = \sqrt{\rho(G^T G)}.$$

First,

$$G^T = \begin{pmatrix} 0 & 0 & 0 & -\frac{1}{2} \\ 0 & 0 & \frac{1}{2} & -\frac{1}{2} \\ -\frac{1}{2} & \frac{1}{2} & 0 & 0 \\ -\frac{1}{2} & 0 & 0 & 0 \end{pmatrix}.$$

Hence

$$G^T G = \begin{pmatrix} \frac{1}{4} & \frac{1}{4} & 0 & 0 \\ \frac{1}{4} & \frac{1}{2} & 0 & 0 \\ 0 & 0 & \frac{1}{2} & \frac{1}{4} \\ 0 & 0 & \frac{1}{4} & \frac{1}{4} \end{pmatrix}.$$

So

$$\lambda I - G^T G = \begin{pmatrix} \lambda - \frac{1}{4} & -\frac{1}{4} & 0 & 0 \\ -\frac{1}{4} & \lambda - \frac{1}{2} & 0 & 0 \\ 0 & 0 & \lambda - \frac{1}{2} & -\frac{1}{4} \\ 0 & 0 & -\frac{1}{4} & \lambda - \frac{1}{4} \end{pmatrix}.$$

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Exercise 1.1(b) – Characteristic polynomial of $G^T G$

We compute

$$\chi_{G^T G}(\lambda) = \det(\lambda I - G^T G).$$

Expanding along the first row,

$$\chi_{G^T G}(\lambda) = \left(\lambda - \frac{1}{4}\right) \begin{vmatrix} \lambda - \frac{1}{2} & 0 & 0 \\ 0 & \lambda - \frac{1}{2} & -\frac{1}{4} \\ 0 & -\frac{1}{4} & \lambda - \frac{1}{4} \end{vmatrix} + \frac{1}{4} \begin{vmatrix} -\frac{1}{4} & 0 & 0 \\ 0 & \lambda - \frac{1}{2} & -\frac{1}{4} \\ 0 & -\frac{1}{4} & \lambda - \frac{1}{4} \end{vmatrix}.$$

This gives

$$\chi_{G^T G}(\lambda) = \left(\lambda - \frac{1}{4}\right) \left(\lambda - \frac{1}{2}\right) \left[\left(\lambda - \frac{1}{2}\right) \left(\lambda - \frac{1}{4}\right) - \frac{1}{16}\right] - \frac{1}{16} \left[\left(\lambda - \frac{1}{2}\right) \left(\lambda - \frac{1}{4}\right) - \frac{1}{16}\right].$$

So

$$\chi_{G^T G}(\lambda) = \left[\left(\lambda - \frac{1}{4}\right) \left(\lambda - \frac{1}{2}\right) - \frac{1}{16}\right]^2 = \left(\lambda^2 - \frac{3}{4}\lambda + \frac{1}{16}\right)^2.$$

Hence the eigenvalues are:

$$\lambda_{1,2,3,4} = \frac{3 + \sqrt{5}}{8}, \frac{3 + \sqrt{5}}{8}, \frac{3 - \sqrt{5}}{8}, \frac{3 - \sqrt{5}}{8}.$$

Therefore

$$\boxed{\rho(G^T G) = \frac{3 + \sqrt{5}}{8}} \quad \Rightarrow \quad \boxed{\|G\|_2 = \sqrt{\frac{3 + \sqrt{5}}{8}} = \frac{1 + \sqrt{5}}{4} \approx 0.8090 < 1.}$$

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Therefore

$$\boxed{\rho(G^T G) = \frac{3 + \sqrt{5}}{8}} \implies \boxed{\|G\|_2 = \sqrt{\frac{3 + \sqrt{5}}{8}} = \frac{1 + \sqrt{5}}{4} \approx 0.8090 < 1.}$$

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$$\rho(G^T G) = \frac{3 + \sqrt{5}}{8}$$

$\Rightarrow$

$$\|G\|_2 = \sqrt{\frac{3 + \sqrt{5}}{8}} = \frac{1 + \sqrt{5}}{4} \approx 0.8090 < 1.$$

Exercise 1.1(b) – Characteristic polynomial of $G^T G$

We compute

$$\chi_{G^T G}(\lambda) = \det(\lambda I - G^T G).$$

Expanding along the first row,

$$\chi_{G^T G}(\lambda) = \left(\lambda - \frac{1}{4}\right) \begin{vmatrix} \lambda - \frac{1}{2} & 0 & 0 \\ 0 & \lambda - \frac{1}{2} & -\frac{1}{4} \\ 0 & -\frac{1}{4} & \lambda - \frac{1}{4} \end{vmatrix} + \frac{1}{4} \begin{vmatrix} -\frac{1}{4} & 0 & 0 \\ 0 & \lambda - \frac{1}{2} & -\frac{1}{4} \\ 0 & -\frac{1}{4} & \lambda - \frac{1}{4} \end{vmatrix}.$$

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So

$$\chi_{G^T G}(\lambda) = \left[\left(\lambda - \frac{1}{4}\right) \left(\lambda - \frac{1}{2}\right) - \frac{1}{16}\right]^2 = \left(\lambda^2 - \frac{3}{4}\lambda + \frac{1}{16}\right)^2.$$

Hence the eigenvalues are:

$$\lambda_{1,2,3,4} = \frac{3 + \sqrt{5}}{8}, \frac{3 + \sqrt{5}}{8}, \frac{3 - \sqrt{5}}{8}, \frac{3 - \sqrt{5}}{8}.$$

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Now we compute $\rho(G)$ from the characteristic polynomial of the full matrix. We write

$$\lambda I - G = \begin{pmatrix} \lambda & 0 & \frac{1}{2} & \frac{1}{2} \\ 0 & \lambda & -\frac{1}{2} & 0 \\ 0 & -\frac{1}{2} & \lambda & 0 \\ \frac{1}{2} & \frac{1}{2} & 0 & \lambda \end{pmatrix}.$$

Hence $\chi_G(\lambda) = \det(\lambda I - G).$

Expanding along the second row,
$$\chi_G(\lambda) = \lambda \begin{vmatrix} \lambda & \frac{1}{2} & \frac{1}{2} \\ -\frac{1}{2} & \lambda & 0 \\ \frac{1}{2} & 0 & \lambda \end{vmatrix} + \frac{1}{2} \begin{vmatrix} \lambda & 0 & \frac{1}{2} \\ 0 & -\frac{1}{2} & 0 \\ \frac{1}{2} & \frac{1}{2} & \lambda \end{vmatrix}.$$

The first 3×3 determinant is
$$\lambda \begin{vmatrix} \lambda & 0 \\ 0 & \lambda \end{vmatrix} - \frac{1}{2} \begin{vmatrix} -\frac{1}{2} & 0 \\ \frac{1}{2} & \lambda \end{vmatrix} + \frac{1}{2} \begin{vmatrix} -\frac{1}{2} & \lambda \\ \frac{1}{2} & 0 \end{vmatrix} = \lambda^3.$$

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Exercise 1.1(b) – Spectral radius of G and convergence

Therefore

$$\chi_G(\lambda) = \lambda \cdot \lambda^3 + \frac{1}{2} \left[-\frac{1}{2} \left(\lambda^2 - \frac{1}{4} \right) \right] = \lambda^4 - \frac{1}{4} \lambda^2 + \frac{1}{16}.$$

So

$$\chi_G(\lambda) = \left(\lambda^2 - \frac{1}{4} \right)^2 = \left(\lambda - \frac{1}{2} \right)^2 \left(\lambda + \frac{1}{2} \right)^2.$$

Therefore the eigenvalues are

$$\lambda = \pm \frac{1}{2} \quad (\text{each with multiplicity } 2).$$

Hence

$$\rho(G) = \frac{1}{2} < 1.$$

By the convergence theorem,

$$\rho(G) < 1 \implies x^{(k+1)} = Gx^{(k)} + c \text{ converges for every } x^{(0)}.$$

Final answer:

$$\|G\|_1 = 1, \quad \|G\|_\infty = 1, \quad \|G\|_2 = \frac{1 + \sqrt{5}}{4}, \quad \rho(G) = \frac{1}{2}.$$

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Exercise 1.1(c) – Compute $x^{(2)}$ for $x^{(0)} = (1, -1, 0, 1)^T$

We use $x^{(k+1)} = Gx^{(k)} + c$.

For $x^{(0)} = \begin{pmatrix} 1 \\ -1 \\ 0 \\ 1 \end{pmatrix},$

we get $x^{(1)} = \begin{pmatrix} 1 - \frac{1}{2}(0) - \frac{1}{2}(1) \\ -1 + \frac{1}{2}(0) \\ \frac{1}{2}(-1) \\ -\frac{1}{2}(1) - \frac{1}{2}(-1) \end{pmatrix} = \begin{pmatrix} \frac{1}{2} \\ -1 \\ -\frac{1}{2} \\ 0 \end{pmatrix}$ and $x^{(2)} = \begin{pmatrix} 1 - \frac{1}{2}(-\frac{1}{2}) - \frac{1}{2}(0) \\ -1 + \frac{1}{2}(-\frac{1}{2}) \\ \frac{1}{2}(-1) \\ -\frac{1}{2}(\frac{1}{2}) - \frac{1}{2}(-1) \end{pmatrix} = \begin{pmatrix} \frac{5}{4} \\ -\frac{5}{4} \\ -\frac{1}{2} \\ \frac{1}{4} \end{pmatrix}$

Hence $x^{(2)} = \left(\frac{5}{4}, -\frac{5}{4}, -\frac{1}{2}, \frac{1}{4} \right)^T.$

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Hence $x^{(2)} = \left(\frac{5}{4}, -\frac{5}{4}, -\frac{1}{2}, \frac{1}{4} \right)^T.$

Exercise 1.2(a) – Which block corresponds to which method?

We compare the convergence speeds through the spectral radius. For Method 1, we already found

$$\rho(G) = \frac{1}{2}.$$

For Gauss-Seidel,

$$\rho(G_{GS}) = \frac{5}{8}.$$

Since

$$\frac{1}{2} < \frac{5}{8},$$

Method 1 must converge *faster* than Gauss-Seidel.

Now look at the two blocks in the table: the one that approaches the exact solution more rapidly must correspond to Method 1. The **lower block** is visibly closer to the exact vector after the same numbers of iterations. Therefore the lower block is the faster method, namely Method 1. So

lower block = Method 1, upper block = Gauss-Seidel.
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Exercise 1.2(a) – Which block corresponds to which method?

We compare the convergence speeds through the spectral radius. For Method 1, we already found

$$\rho(G) = \frac{1}{2}.$$

For Gauss-Seidel,

$$\rho(G_{GS}) = \frac{5}{8}.$$

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Exercise 1.2(b)(i) – Residual test for GS

We check $\|b - Ax^{(k)}\|_2 \leq \varepsilon, \quad k = 5, \quad \varepsilon = 10^{-5}.$

For Gauss-Seidel, $x_{GS}^{(5)} = \begin{pmatrix} \frac{9881}{8192} \\ -\frac{9881}{8192} \\ -\frac{9881}{16384} \\ 0 \end{pmatrix} \Rightarrow r^{(5)} = b - Ax_{GS}^{(5)} = \begin{pmatrix} \frac{3125}{32768} \\ 0 \\ 0 \\ 0 \end{pmatrix}.$

Hence $\|r^{(5)}\|_2 = \frac{3125}{32768} \approx 9.54 \times 10^{-2} 10^{-5}.$

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the stopping criterion is not satisfied.

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Exercise 1.2(b)(ii) – Relative difference test for Method 1

We check

$$\frac{\|x^{(k)} - x^{(k-1)}\|_{\infty}}{\|x^{(k)}\|_{\infty}} \leq \varepsilon, \quad k = 5, \quad \varepsilon = 10^{-1}.$$

For Method 1,

$$x^{(4)} = \begin{pmatrix} \frac{5}{4} \\ -\frac{5}{4} \\ -\frac{5}{8} \\ 0 \end{pmatrix}, \quad x^{(5)} = \begin{pmatrix} \frac{21}{16} \\ -\frac{21}{16} \\ -\frac{5}{8} \\ 0 \end{pmatrix} \Rightarrow x^{(5)} - x^{(4)} = \begin{pmatrix} \frac{1}{16} \\ -\frac{1}{16} \\ 0 \\ 0 \end{pmatrix}, \quad \|x^{(5)} - x^{(4)}\|_{\infty} = \frac{1}{16}.$$

$$\text{Also, } \|x^{(5)}\|_{\infty} = \frac{21}{16}.$$

$$\text{Therefore } \frac{\|x^{(5)} - x^{(4)}\|_{\infty}}{\|x^{(5)}\|_{\infty}} = \frac{1/16}{21/16} = \frac{1}{21} \approx 4.76 \times 10^{-2} < 10^{-1}.$$

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Exercise 1.2(b)(iii) – Exact error test for GS

We check

$$\|x^{(k)} - x^*\|_1 \leq \varepsilon, \quad k = 4, \quad \varepsilon = 10^{-3}.$$

For Gauss-Seidel,

$$x_{GS}^{(4)} = \begin{pmatrix} \frac{1157}{1024} \\ -\frac{1157}{1024} \\ -\frac{1157}{2048} \\ 0 \end{pmatrix}, \quad x^* = \begin{pmatrix} \frac{4}{3} \\ -\frac{4}{3} \\ -\frac{2}{3} \\ 0 \end{pmatrix} \Rightarrow x_{GS}^{(4)} - x^* = \begin{pmatrix} -\frac{625}{3072} \\ \frac{625}{3072} \\ \frac{625}{6144} \\ 0 \end{pmatrix}.$$

Hence
$$\|x_{GS}^{(4)} - x^*\|_1 = \frac{625}{3072} + \frac{625}{3072} + \frac{625}{6144} = \frac{3125}{6144} \approx 5.086 \times 10^{-1} > 10^{-3}.$$

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Exercise 1.3 – Jacobi formulas

For Jacobi, with $A = D - (E + F)$, we use only old components:

$$\begin{cases} x_1^{(k+1)} = \frac{1 - x_2^{(k)} - \frac{1}{2}x_3^{(k)} - x_4^{(k)}}{2}, \\ x_2^{(k+1)} = -x_1^{(k)} - \frac{1}{2}x_4^{(k)}, \\ x_3^{(k+1)} = -\frac{1}{2}x_1^{(k)} - x_4^{(k)}, \\ x_4^{(k+1)} = -\frac{x_1^{(k)} + \frac{1}{2}x_2^{(k)} + x_3^{(k)}}{2}. \end{cases}$$

We start from

$$x^{(0)} = (0, 0, 0, 0)^T.$$

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Exercise 1.3 – Two Jacobi iterations

First iteration:

$$x_J^{(1)} = \begin{pmatrix} \frac{1}{2} \\ 0 \\ 0 \\ 0 \end{pmatrix}.$$

Second iteration:

$$x_J^{(2)} = \begin{pmatrix} \frac{1 - 0 - 0 - 0}{2} \\ -\frac{1}{2} - \frac{1}{2}(0) \\ -\frac{1}{2} \cdot \frac{1}{2} - 0 \\ -\frac{\frac{1}{2} + 0 + 0}{2} \end{pmatrix} = \begin{pmatrix} \frac{1}{2} \\ -\frac{1}{2} \\ -\frac{1}{4} \\ -\frac{1}{4} \end{pmatrix}.$$

Therefore

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For Gauss-Seidel, we reuse the newest values immediately:

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Again, we start from

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Exercise 1.3 – Two Gauss-Seidel iterations

First iteration:

$$x_{GS}^{(1)} = \begin{pmatrix} \frac{1}{2} \\ -\frac{1}{2} \\ -\frac{1}{4} \\ 0 \end{pmatrix}.$$

Second iteration:

$$x_1^{(2)} = \frac{1 - (-\frac{1}{2}) - \frac{1}{2}(-\frac{1}{4}) - 0}{2} = \frac{13}{16},$$

$$x_2^{(2)} = -\frac{13}{16} - \frac{1}{2} \cdot 0 = -\frac{13}{16}, \quad x_3^{(2)} = -\frac{1}{2} \cdot \frac{13}{16} - 0 = -\frac{13}{32},$$

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Exercise 1.3 – Errors after two iterations

Recall the exact solution

$$x^* = \left(\frac{4}{3}, -\frac{4}{3}, -\frac{2}{3}, 0 \right)^T.$$

For Jacobi,

$$x_J^{(2)} - x^* = \begin{pmatrix} -\frac{5}{6} \\ \frac{5}{6} \\ \frac{5}{12} \\ -\frac{1}{4} \end{pmatrix} \Rightarrow \boxed{\|x_J^{(2)} - x^*\|_1 = \frac{7}{3} \approx 2.3333.}$$

For Gauss-Seidel,

$$x_{GS}^{(2)} - x^* = \begin{pmatrix} -\frac{25}{48} \\ \frac{25}{48} \\ \frac{25}{96} \\ 0 \end{pmatrix} \Rightarrow \boxed{\|x_{GS}^{(2)} - x^*\|_1 = \frac{125}{96} \approx 1.3021.}$$

So after two iterations, **Gauss-Seidel is already more accurate than Jacobi.**

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$$x_J^{(2)} - x^* = \begin{pmatrix} -\frac{5}{6} \\ \frac{5}{6} \\ \frac{5}{12} \\ -\frac{1}{4} \end{pmatrix} \Rightarrow \boxed{\|x_J^{(2)} - x^*\|_1 = \frac{7}{3} \approx 2.3333.}$$

For Gauss-Seidel,

$$x_{GS}^{(2)} - x^* = \begin{pmatrix} -\frac{25}{48} \\ \frac{25}{48} \\ \frac{25}{96} \\ 0 \end{pmatrix} \Rightarrow \boxed{\|x_{GS}^{(2)} - x^*\|_1 = \frac{125}{96} \approx 1.3021.}$$

So after two iterations, **Gauss-Seidel is already more accurate than Jacobi.**

Exercise 1.3 – Errors after two iterations

Recall the exact solution

$$x^* = \left(\frac{4}{3}, -\frac{4}{3}, -\frac{2}{3}, 0 \right)^T.$$

For Jacobi,

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Exercise 2 – Matrix and objective

We consider

$$A = \begin{pmatrix} 1 & 0 & \beta \\ \alpha & 1 & \beta \\ -\beta & \beta & 1 \end{pmatrix}.$$

We want to determine:

- 1 the domain D_1 where A is strictly diagonally dominant,
- 2 the exact convergence domain D_2 for Jacobi,
- 3 the exact convergence domain D_3 for Gauss-Seidel,
- 4 and compare these three sets.

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Exercise 2.1 – Strict diagonal dominance

A matrix is strictly diagonally dominant by rows if

$$|a_{ii}| > \sum_{j \neq i} |a_{ij}| \quad \text{for every row } i.$$

Here all diagonal entries are equal to 1.

Row 1 gives

$$1 > |0| + |\beta| \implies |\beta| < 1.$$

Row 2 gives

$$1 > |\alpha| + |\beta|.$$

Row 3 gives

$$1 > |-\beta| + |\beta| = 2|\beta| \implies |\beta| < \frac{1}{2}.$$

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Exercise 2.1 – Domain D_1 and conclusion

Combining the three inequalities,

$$D_1 = \left\{ (\alpha, \beta) \in \mathbb{R}^2 : |\beta| < 1, |\alpha| + |\beta| < 1, 2|\beta| < 1 \right\}.$$

Since $2|\beta| < 1$ already implies $|\beta| < 1$, we simplify to

$$D_1 = \left\{ (\alpha, \beta) \in \mathbb{R}^2 : |\beta| < \frac{1}{2}, |\alpha| + |\beta| < 1 \right\}.$$

By the sufficient convergence theorem,

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Exercise 2.2 – Goal

We want to determine:

$$D_2 = \{(\alpha, \beta) \in \mathbb{R}^2 \mid \text{Jacobi converges}\}, \quad D_3 = \{(\alpha, \beta) \in \mathbb{R}^2 \mid \text{Gauss-Seidel converges}\}.$$

We consider

$$A = \begin{pmatrix} 1 & 0 & \beta \\ \alpha & 1 & \beta \\ -\beta & \beta & 1 \end{pmatrix}.$$

We will:

- 1 write $A = D - E - F$,
- 2 compute the iteration matrix for Jacobi,
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Exercise 2.2 – Decomposition $A = D - E - F$

Recall that

$$A = D - E - F,$$

where

$$D = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \quad E = \begin{pmatrix} 0 & 0 & 0 \\ -\alpha & 0 & 0 \\ \beta & -\beta & 0 \end{pmatrix}, \quad F = \begin{pmatrix} 0 & 0 & -\beta \\ 0 & 0 & -\beta \\ 0 & 0 & 0 \end{pmatrix}.$$

Indeed,

$$D - E - F = \begin{pmatrix} 1 & 0 & \beta \\ \alpha & 1 & \beta \\ -\beta & \beta & 1 \end{pmatrix} = A.$$

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Exercise 2.2 – Jacobi iteration matrix

For Jacobi, the iteration matrix is

$$B_J = D^{-1}(E + F).$$

Since here $D = I_3$, we have $D^{-1} = I_3$, hence

$$B_J = E + F.$$

Therefore,

$$B_J = \begin{pmatrix} 0 & 0 & -\beta \\ -\alpha & 0 & -\beta \\ \beta & -\beta & 0 \end{pmatrix}.$$

So the Jacobi method is

$$x^{(k+1)} = B_J x^{(k)} + D^{-1}b.$$

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Exercise 2.2 – Characteristic polynomial of B_J

We compute

$$\chi_{B_J}(\lambda) = \det(\lambda I - B_J).$$

We have

$$\lambda I - B_J = \begin{pmatrix} \lambda & 0 & \beta \\ \alpha & \lambda & \beta \\ -\beta & \beta & \lambda \end{pmatrix}.$$

Expanding along the first row,

$$\chi_{B_J}(\lambda) = \lambda \begin{vmatrix} \lambda & \beta \\ \beta & \lambda \end{vmatrix} + \beta \begin{vmatrix} \alpha & \lambda \\ -\beta & \beta \end{vmatrix}.$$

Thus

$$\chi_{B_J}(\lambda) = \lambda(\lambda^2 - \beta^2) + \beta(\alpha\beta + \lambda\beta) = \lambda^3 + \alpha\beta^2.$$

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Exercise 2.2 – Spectral radius for Jacobi

Hence all eigenvalues have modulus

$$|\lambda| = |\alpha\beta^2|^{1/3}.$$

So the spectral radius is

$$\rho(B_J) = |\alpha\beta^2|^{1/3}.$$

Therefore, Jacobi converges (for all data) if and only if

$$\rho(B_J) < 1 \iff |\alpha\beta^2|^{1/3} < 1 \iff |\alpha|\beta^2 < 1.$$

Hence

$$D_2 = \left\{ (\alpha, \beta) \in \mathbb{R}^2 \mid |\alpha|\beta^2 < 1 \right\}.$$

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Exercise 2.2 – Gauss-Seidel iteration matrix

For Gauss-Seidel, the iteration matrix is

$$B_{GS} = (D - E)^{-1}F.$$

First,

$$D - E = \begin{pmatrix} 1 & 0 & 0 \\ \alpha & 1 & 0 \\ -\beta & \beta & 1 \end{pmatrix}.$$

Now we compute $(D - E)^{-1}$. we obtain

$$(D - E)^{-1} = \begin{pmatrix} 1 & 0 & 0 \\ -\alpha & 1 & 0 \\ \beta(1 + \alpha) & -\beta & 1 \end{pmatrix}.$$

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Exercise 2.2 – Compute B_{GS}

Now

$$F = \begin{pmatrix} 0 & 0 & -\beta \\ 0 & 0 & -\beta \\ 0 & 0 & 0 \end{pmatrix}.$$

Therefore

$$B_{GS} = (D - E)^{-1}F = \begin{pmatrix} 1 & 0 & 0 \\ -\alpha & 1 & 0 \\ \beta(1 + \alpha) & -\beta & 1 \end{pmatrix} \begin{pmatrix} 0 & 0 & -\beta \\ 0 & 0 & -\beta \\ 0 & 0 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 0 & -\beta \\ 0 & 0 & \beta(\alpha - 1) \\ 0 & 0 & -\alpha\beta^2 \end{pmatrix}.$$

So the Gauss-Seidel method is

$$x^{(k+1)} = B_{GS}x^{(k)} + (D - E)^{-1}b.$$

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Exercise 2.2 – Spectral radius for Gauss-Seidel

The matrix B_{GS} is upper triangular, so its eigenvalues are the diagonal entries:

$$0, \quad 0, \quad -\alpha\beta^2.$$

Hence

$$\rho(B_{GS}) = |\alpha\beta^2|.$$

Therefore, Gauss-Seidel converges if and only if

$$\rho(B_{GS}) < 1 \iff |\alpha\beta^2| < 1.$$

So

$$D_3 = \left\{ (\alpha, \beta) \in \mathbb{R}^2 \mid |\alpha|\beta^2 < 1 \right\}.$$

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Exercise 2.2 – Comparison of D_1, D_2, D_3

We found

$$D_2 = \left\{ (\alpha, \beta) \in \mathbb{R}^2 \mid |\alpha|\beta^2 < 1 \right\}, \quad D_3 = \left\{ (\alpha, \beta) \in \mathbb{R}^2 \mid |\alpha|\beta^2 < 1 \right\}.$$

Thus

$$D_2 = D_3.$$

From Question 1,

$$D_1 = \left\{ (\alpha, \beta) \in \mathbb{R}^2 \mid |\beta| < \frac{1}{2}, |\alpha| + |\beta| < 1 \right\}.$$

Hence

$$D_1 \subseteq D_2 = D_3.$$

Exercise 2.2 – Comparison of D_1, D_2, D_3

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$$D_2 = \left\{ (\alpha, \beta) \in \mathbb{R}^2 \mid |\alpha|\beta^2 < 1 \right\}, \quad D_3 = \left\{ (\alpha, \beta) \in \mathbb{R}^2 \mid |\alpha|\beta^2 < 1 \right\}.$$

Thus

$$D_2 = D_3.$$

From Question 1,

$$D_1 = \left\{ (\alpha, \beta) \in \mathbb{R}^2 \mid |\beta| < \frac{1}{2}, |\alpha| + |\beta| < 1 \right\}.$$

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